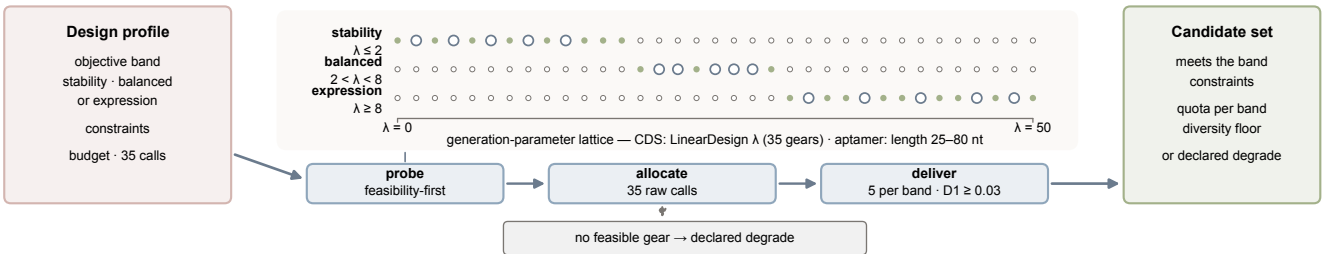
GenNav · constraint navigation
Layer-2 generation step · mRNA CDS · aptamer
feasible
infeasible
delivered
Design profile
objective band
stability · balanced
or expression
constraints
budget · 35 calls
stability
λ ≤ 2
balanced
2 < λ < 8
expression
λ ≥ 8
λ = 0
λ = 50
generation-parameter lattice — CDS: LinearDesign λ (35 gears) · aptamer: length 25–80 nt
probe
feasibility-first
allocate
35 raw calls
deliver
5 per band · D1 ≥ 0.03
no feasible gear → declared degrade
Candidate set
meets the band
constraints
quota per band
diversity floor
or declared degrade